Experiment No-17 : Sum of the Integers from 1 to n (Prolog)

Aim

To write a Prolog program to find the sum of the integers from 1 to n.

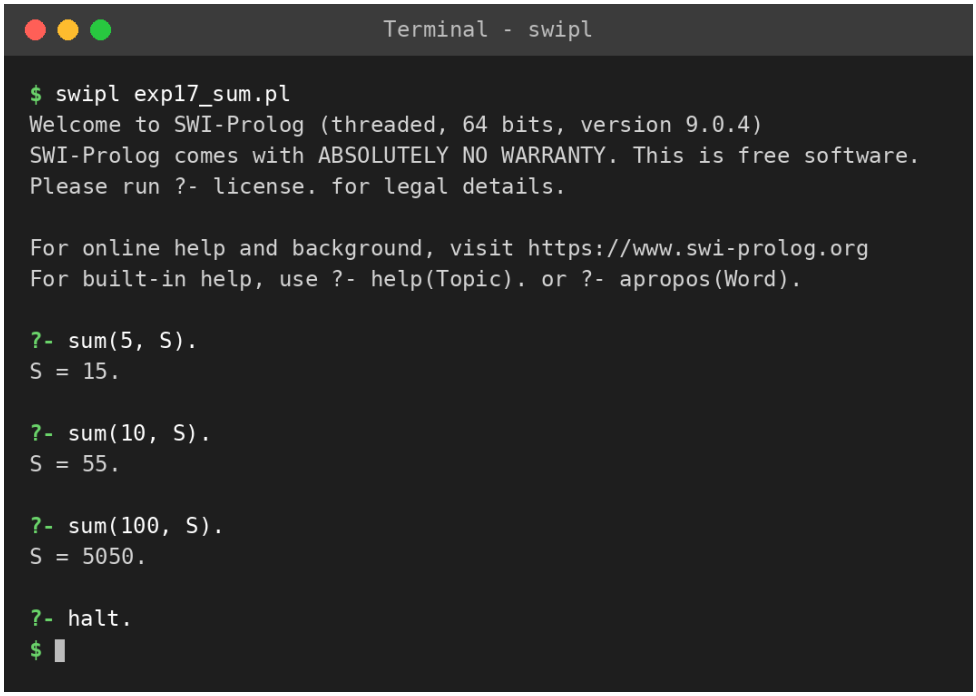
Programming Code:

```
% Sum of the integers from 1 to N

% Base case: sum of integers up to 0 is 0
sum(0, 0) :- !.

% Recursive case: sum(N) = N + sum(N-1)
sum(N, S) :-
    N > 0,
    N1 is N - 1,
    sum(N1, S1),
    S is S1 + N.
```

OUTPUT:



Expected Output

- The query sum(5, S). gives S = 15, which is 1 + 2 + 3 + 4 + 5.
- The query sum(10, S). gives S = 55.
- The query sum(100, S). gives S = 5050.
- Each query returns a single answer ending with a full stop, because the cut (!) in the base case stops Prolog from searching for more solutions.
- The recursion reduces n by 1 at each call until the base case sum(0, 0) is reached.

Result

Thus, the Prolog program to find the sum of the integers from 1 to n was executed successfully using recursion with a base case and a recursive rule.